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YE XU, Mid Sweden University, Sundsvall, Sweden

SEBASTIAN BADER, Mid Sweden University, Sundsvall, Sweden

BENGT OELMANN, Mid Sweden University, Sundsvall, Sweden

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Ye Xu
Mid Sweden University
Sundsvall, Sweden
ye.xu@miun.se

Sebastian Bader
Mid Sweden University
Sundsvall, Sweden
sebastian.bader@miun.se

Bengt Oelmann
Mid Sweden University
Sundsvall, Sweden
bengt.oelmann@miun.se

ABSTRACT

The feasibility of energy harvesting as a viable alternative for powering low-energy electronics has been demonstrated through advancements in transduction mechanisms. Energy harvesters incorporating counterweights have gained attention in rotational energy harvesting to develop single-anchored devices with flexible placement and easy installation. In this work, a three-phase variable reluctance energy harvester (VREH) with low torque ripple is combined with a counterweight to facilitate a single-anchored design, specifically targeting low rotational speed applications. The energy harvester is integrated with a low-power sensor system to enable energy-neutral operation. We present the design, implementation, and evaluation of an on-rotor RPM sensor system powered by the single-anchored three-phase VREH. Experimental evaluations on a laboratory test bench demonstrate the system performance under varying conditions, with the ability to supply the sensor system at low speeds achieving, for example, a 3.5 Hz sample rate at a low speed of 3 rpm. Evaluations of the system illustrate that pendulum effects induced by the interaction of the cogging torque and the gravitational torque improve the output power of the harvester under low-speed conditions. This promises for the proposed design to be suitable to power wireless sensors for industrial condition monitoring, providing a flexible solution for energy-neutral sensor systems with reduced installation complexity.

CCS CONCEPTS

• **Hardware** → **Renewable energy**; *Wireless devices*; **Power estimation and optimization**.

KEYWORDS

energy harvesting, electromagnetic transduction, pendulum effect, RPM sensor, self-powered wireless sensor, variable reluctance

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1 INTRODUCTION

Energy harvesting research concentrates on the extraction of energy from ambient sources and its effective conversion into electrical energy. This includes the conversion of rotational energy into electrical energy for the supply of low-power electronics, such as Internet of Things (IoT) devices and wireless sensor systems [3, 19]. Rotational components in industrial applications require monitoring, which is enabled by modern low-power sensing techniques. These techniques use, for example, inertial measurement units mounted directly on the rotating objects to obtain information such as rotational speed or component condition [2]. Energy harvesting from mechanical rotation has grown in attraction to supply these sensing systems, promising to reduce dependence on conventional batteries for low-energy electronic devices [1, 3, 5, 6, 10].

Among the various mechanisms of rotational energy harvesting, electromagnetic approaches are a desirable alternative, due to their robustness, power density, and low-cost. Electromagnetic rotational energy harvesters typically require two anchor points, namely one for the stator component and one for the rotor component [5]. However, it is possible to employ a counterweight that due to gravity functions as a friction pendulum [21]. Utilizing a counterweight enables the development of a single-anchor structure, while maintaining relative motion between coils and magnets, as is typical in electromagnetic energy harvesting [12, 13, 21]. The absence of a fixed stator component simplifies installation and facilitates use in applications where access to a static attachment point is not possible.

Variable reluctance energy harvesters (VREHs) have been demonstrated as attractive solutions for rotational applications due to their low complexity and robustness [4, 7, 11, 14, 20]. Nevertheless, VREHs also require relative motion between a pickup unit and a toothed wheel and thus have so far only been investigated as solutions with two anchor points. Furthermore, as presented in [16, 17], VREHs induce a strong torque ripple, which may necessitate the use of a large mass if a gravity-supported stator with counterweight is to be employed. In [18], it was shown that this torque ripple can be minimized through a three-phase design of the pickup unit.

To address the aforementioned challenges, in this study we present a proof-of-concept for an on-rotor RPM sensor system powered by a single-anchor, three-phase VREH. The VREH uses electromagnetic induction to generate energy from the relative motion between the rotor and a gravity-supported stator. The sensor system measures the rotary object's angular velocity using a micro-electromechanical system (MEMS) gyroscope and transmits the acquired data wirelessly using a Bluetooth low energy (BLE) transceiver. In experimental evaluations on a 400 mm target rotating at rates of 3 rpm to 17 rpm, we investigate the system performance

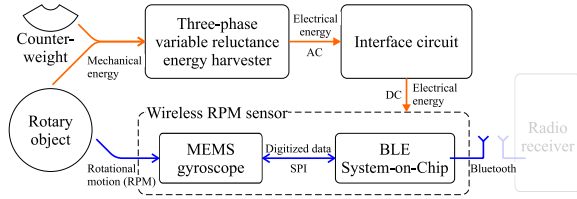


Figure 1: Overview of the on-rotor RPM sensor design with its key building blocks.

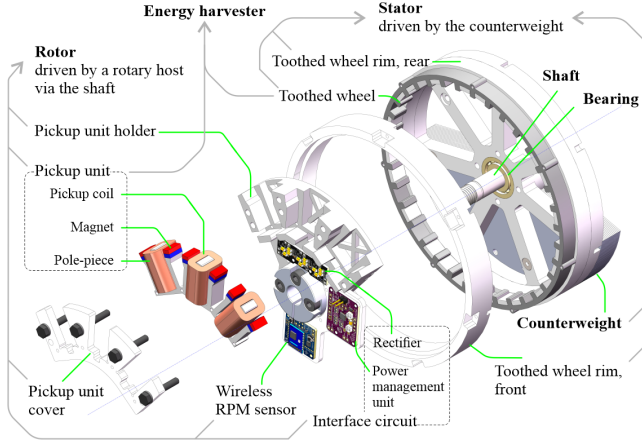


Figure 2: Exploded view of the self-powered RPM sensor system implementation.

under varying conditions, including influences of rotational speed and mounting location. The results show that the three-phase VREH already at 3 rpm provides a power of 200 μ W, which is sufficient to enable the sensor system to operate at a 3.5 Hz sample rate.

The study demonstrates that the proposed design achieves a single-anchor rotational speed monitoring system integrating sensing and power supply mechanisms. The single-anchored design of the VREH allows it to be mounted on a rotary object regardless of location, achieving flexible and low-cost installations. The investigated solution is particularly suitable for applications involving large-diameter rotating objects with low rotational speeds, such as feeders, drums, roll mills typically found in industrial applications [8, 9], where it can enable energy-autonomous condition monitoring.

2 SYSTEM DESIGN AND IMPLEMENTATION

The key components of the proposed system are shown in Fig. 1, which follows a typical wireless sensor node architecture with control, radio, and sensing units. It also integrates the variable reluctance energy harvester and an interface circuit for rectification and power regulation.

Figure 2 illustrates the implemented system to be fixed on a rotating host. The system integrates the rotating pickup unit and sensor system into the gravity-supported stator. The rotor is driven by the monitored host via a shaft, while the stator remains stationary due

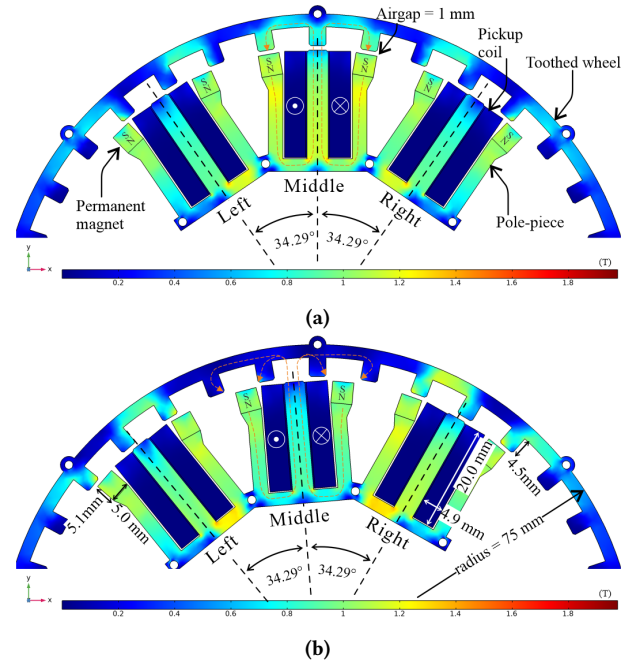


Figure 3: Structure of a three-phase VREH with magnetic flux density distribution in (a) aligned and (b) unaligned position. Primary flux directions are indicated by dashed lines for the central pickup unit.

to the gravitational force acting on the counterweight. A bearing is used to enable relative rotor-stator motion.

2.1 Three-phase VREH

The VREH utilizes relative motion between a pickup unit, containing coils and magnets, and a ferromagnetic toothed wheel. The change in relative position between the pickup unit and the teeth creates a magnetic reluctance variation, and thus an electromagnetic induction. In a three-phase VREH, multiple pickup units are arranged at a specific distance to each other, which affects output power and torque.

Figure 3 depicts the simulated magnetic flux distribution at the VREHs two extreme positions (with respect to the center pickup unit). When the pickup unit is aligned with the teeth (Fig. 3a), magnetic flux is maximized due to a minimal airgap. In the unaligned position (Fig. 3b), the airgap is maximized and thus the magnetic flux is reduced. During rotation, the system undergoes periodical shifts between the aligned and unaligned positions, resulting in an alternating magnetic flux. The magnetic flux change leads to an electromagnetic induction in the coil, generating an alternating voltage according to Faraday's law.

The power extraction capabilities of the energy harvester may be determined by applying the maximum power transfer theorem. Given that the VREH operates at low angular velocities, it can be assumed that the inductive component of the pickup coil is negligible. Thus, with resistive impedance matching ($R_{Coil} = R_{Load}$),

the maximum power at each pickup coil can be estimated as

$$P_{\max} = \frac{1}{4R_{\text{Coil}}} (-\omega N_{\text{Coil}} \frac{d\phi}{d\theta})^2, \quad (1)$$

where ω is the rotational speed in rad/s, N_{Coil} denotes the number of turns of the pickup coil, ϕ is the flux linkage in the pickup coil, and θ is the mechanical angular displacement of the toothed wheel. The frequency of the output voltages is expressed by

$$f = \omega N_T, \quad (2)$$

where N_T is the number of teeth of the toothed wheel.

The interaction between the pickup unit and the teeth also leads to a cogging torque. In VREHs, cogging torque is a major component contributing to torque ripple. Significant torque ripple can result in issues such as vibrations, noise, and unstable motion in rotary hosts, especially in applications demanding smooth operation at low speeds and light loads [18]. The torque of a harvester with multiple pickup units is described as the combination of each pickup unit's torque, which can be represented by a Fourier series in amplitude-phase form [18]:

$$\tau = \sum_{p=0}^{N_{\text{pickup}}-1} \left\{ \sum_{k=1}^{\infty} \tau_{N_{\text{pickup}}, N_T, k} \cdot \sin[N_T \cdot k \cdot (\theta - p \cdot \beta)] \right\}, \quad (3)$$

$k \in \mathbb{Z}^+, N_{\text{pickup}} = 3,$

where p indicates the p -th pickup unit, k is the harmonic order, β is the angle that each pickup unit is shifted relative to the others, and $\tau_{N_{\text{pickup}}, N_T, k}$ is the cogging torque for each pickup unit as described in [18].

This reveals that changing β allows pickup units to alter the cogging torque waveform through symmetrical and asymmetrical patterns in the trigonometric term. An optimized value of β can mitigate cogging torque via compensation effects. In this manner, the three pickup units are optimized for minimal torque, resulting in an angular distance of 34.29° as shown in Fig. 3. The numerical simulation indicates that the resulting torque ripple has a peak value of only 0.012 N m, as compared to 0.46 N m for a single pickup unit. The three-phase VREH generates sinusoidal voltage waveforms with a frequency of f that are phase-shifted by 120 electrical degrees to each other.

For the implementation of the three-phase VREH, the pole-piece and toothed wheel are constructed of laminated electrical steel (NSC-35H230) with 28 layers and a total depth of 10.15 mm. Two permanent N50-NdFeB magnets are used in each pickup unit to provide a consistent magnetomotive force (MMF). The pickup coils are wound on the center poles of the pole-piece, resulting in an inductance of 1.2 mH for each coil.

2.2 Counterweight Design

In order to allow the VREH to operate with a single anchor point, a counterweight is integrated into the stator component of the harvester. Incorporating a counterweight introduces a gravitational torque, effectively maintaining the toothed wheel stationary in relation to the pickup unit as depicted in Fig. 4. However, a number of aspects need to be considered in the design.

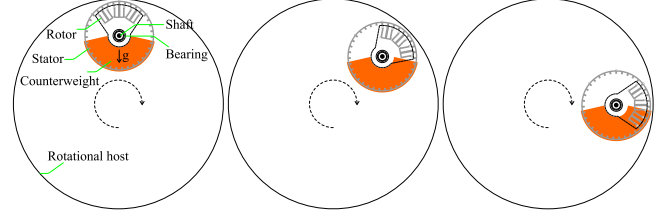


Figure 4: Conceptual principle of the single-anchor VREH with counterweight.

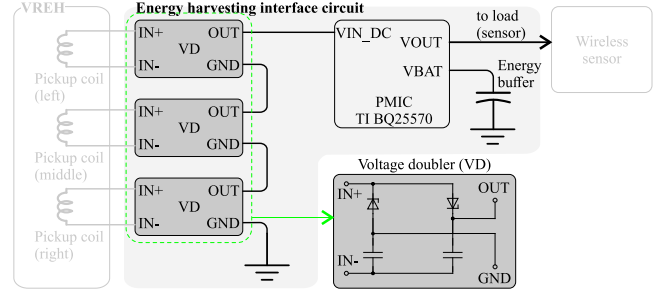


Figure 5: Schematic overview of the interface circuit, containing three voltage doublers connected in series and a power management circuit based on a Texas Instruments BQ25570.

A critical angular velocity exists for a given counterweight mass, at which the magnetic torque outweighs the torque due to gravity and the counterweight rotates along with the pickup unit. As a result, the toothed wheel and the pickup unit will no longer have a relative displacement and thus no power will be generated. A minimum counterweight mass should thus be selected for each application.

The VREH's torque ripple can also exceed the gravitational force temporarily, which can lead to the counterweight following the pickup unit for some time. However, at a particular angular position, where the gravitational torque outweighs the torque ripple again, the counterweight and toothed wheel disengage from the pickup unit's driving force. At this moment, they rapidly rotate downwards until they are stopped by a sufficiently high counter-torque. Consequently, this action induces a pendulum-like oscillatory motion in the counterweight due to inertia. This effect can be exploited at low rotational speeds to increase the harvested energy, as the fast downwards motion increases the magnetic field change in the VREH.

For the pendulum-effect to generate most benefit, the counterweight should be as small as possible, while avoiding the stator to continuously rotate with the pickup unit.

2.3 Energy Harvesting Interface Circuit

The interface circuit for the VREH is shown in Fig. 5 and consists of two components. A rectifier circuit is used to convert the alternating current (AC) of the VREH to a direct current (DC) for the sensing system. Moreover, a DC-DC power conditioning circuit is used for effective energy storage management and voltage level regulation.

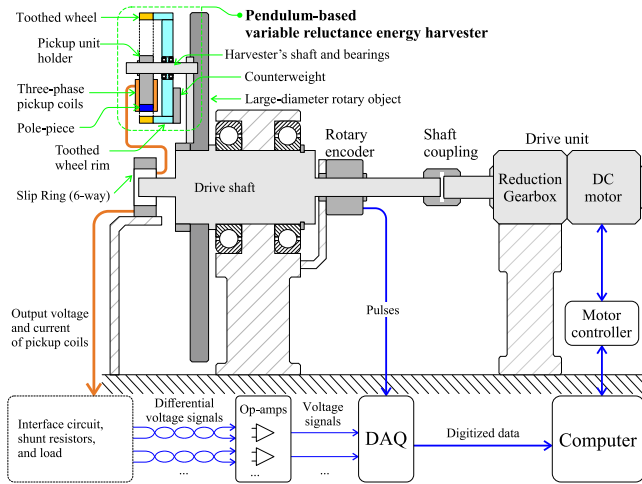


Figure 6: Schematic overview of the experimental setup.

The rectifier circuit comprises three full-wave voltage doublers (VDs) connected in series, each rectifying the output of one pickup coil. At low rotational speeds, VD-based rectifiers demonstrate superior efficiency and output power [15]. Figure 5 illustrates the configuration of the VDs with two diodes and two capacitors for each rectification stage. The output voltage between *OUT* and *GND* is ideally twice the input AC peak voltage between *IN+* and *IN-*.

For the power conditioning circuit, a commercial DC-DC power management integrated circuit (PMIC) (Texas Instruments BQ25570) is utilized. This PMIC is characterized by a low quiescent current of under 500 nA and a high efficiency exceeding 90 %.

2.4 Wireless RPM Sensor

The RPM sensor in the system employs a MEMS gyroscope for rotational speed measurement. MEMS gyroscopes offer cost-efficiency, compact size, and low power consumption due to their integration and mass production. They can also outperform traditional methods, such as optical or magnetic encoders, for larger shaft diameters and lower rotation rates [9]. In this study, a Bosch Sensortec BMI270 was chosen due to its small footprint (2.5 mm × 3.0 mm) and low-power mode current consumption (400 μA). The gyroscope is interfaced through SPI with a Nordic Semiconductor nRF52832 System-on-Chip, integrating an ARM Cortex-M4 microcontroller and BLE transceiver. For the proof-of-concept, digitized angular velocity values are periodically sampled from the gyroscope and wirelessly transmitted to a Bluetooth receiver.

3 EXPERIMENTAL SETUP

Experiments are performed on a rotation setup with a diameter of 400 mm, as shown in Fig. 6. The setup is rotating at speeds ranging from 3 rpm to 17 rpm, driven via a drive shaft by a DC motor and a reduction gearbox. A rotary encoder generates a pulsed output signal for reference. This signal is used to synchronize a data acquisition unit (DAQ) monitoring the rotary object's rotational speed and counting its mechanical rotations. The pickup unit of the energy harvester is mechanical to rotate synchronously with the

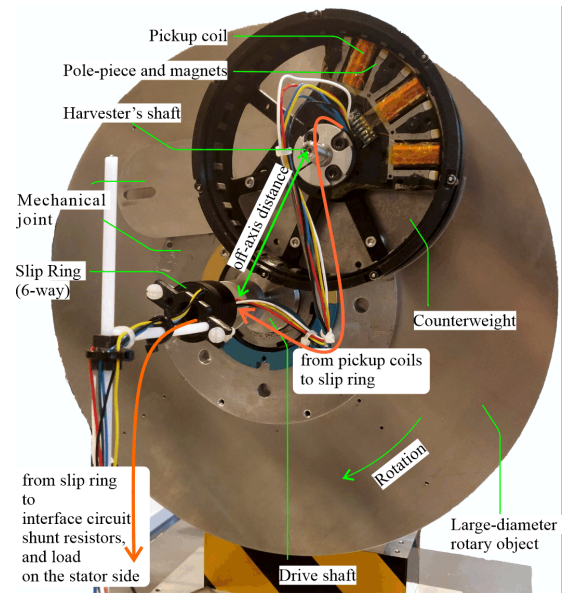


Figure 7: Front view of the experimental setup with the harvester positioned 100 mm off the axis of rotation.

host. For simplified instrumentation during the characterization of the harvester, the interface circuit is not mounted on the rotating setup, but is wired to the harvester using a six-way slip ring, as shown in Fig. 7.

Both the input and output power of the interface circuit are measured in order to assess its performance. Voltages and currents, including the rectifier and PMIC input and output, are digitized using the DAQ system composed of two Analog Devices ADAS3022 circuits. Each ADAS3022 has eight 16-bit ADC channels, accepting analog input signals within the range of -10.24 V to 10.24 V. The signals are sampled at a rate of 10 kHz. Before the digitization, input and output voltages are measured via differential amplifiers employing an INA105 differential-to-single-ended amplifier, while currents are transformed into voltage signals using current sense amplifiers consisting of an INA114 amplifier and a 1-ohm shunt resistor. Prior to all measurements, the amplifier circuits and the DAQ are calibrated using a precision source-measure unit (Keysight B2901A).

4 SYSTEM EVALUATION

4.1 System-level Energy Consumption

The energy consumption of the wireless sensor is a critical factor that determines the amount of energy that needs to be generated in order to ensure energy-neutral system operation. Maintaining a constant 1.75 V supply voltage, the energy consumption can be estimated via the current profile of the node. The current profile is acquired by measuring the voltage drop over a low-side 1 Ω shunt resistor using an INA114 amplifier operating in differential mode with the help of an DAQ. To accurately track the current draw changes, the acquisition was done at a 100 kHz sample rate.

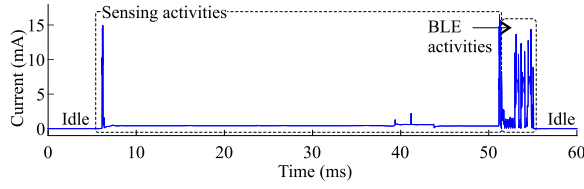


Figure 8: Current profile during the acquisition of one sample on the wireless sensor operating at a supply voltage of 1.75 V.

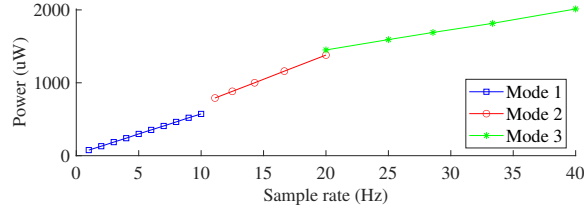


Figure 9: Power requirements of the wireless sensor operating at three different modes with a supply voltage of 1.75 V.

Fig. 8 depicts the current profile of a single wireless sensor sample acquisition. The acquisition process includes gyroscope sampling and readout, as well as wireless transfer of the sampled value, and lasts 49 ms. The required energy for one acquisition is approximately 60 μ J, with sensing activities comprising the largest part (approx. 66 %). With an idle current draw of approximately 8.50 μ A, average power consumption levels for different sample intervals can be calculated. A sample rate of 1 Hz, for example, results in an average power consumption of approximately 75 μ W.

The wireless sensor node can be operated in various modes, determined by the duty-cycling of its two subsystems (the BLE SoC and gyroscope). Fig. 9 illustrates how these modes affect both the system's maximum sample rate and average power consumption. In Mode 1, both the sensing and BLE activities are periodically activated and turned off. The BLE operates in a non-connectable advertising mode with intervals of at least 100 ms. As a result, the maximum achievable sample rate in this mode is limited to 10 Hz, with a total power consumption of up to 570 μ W. To increase the sample rate further, Mode 2 employs a connectable advertisement mode for the BLE, while the gyroscope maintains duty-cycling. In this mode, the gyroscope's initialization period of 45 ms limits the maximum sample rate. Together with an BLE active period of approximately 5 ms, sample rates of up to 20 Hz can be achieved. In Mode 3, the gyroscope remains always-on to avoid re-initialization, while the BLE continues to be duty-cycled. With a minimal BLE advertisement interval of 25 ms, this mode allows for the highest possible sample rate of 40 Hz. To further increase the sample rate, the BLE would need to connect to a receiver. As the connection consumes significant energy itself, this operating mode is not considered in this proof-of-concept.

4.2 Energy Harvesting System Performance

The performance of the harvesting system, which includes output power and overall power conversion efficiency, depends on

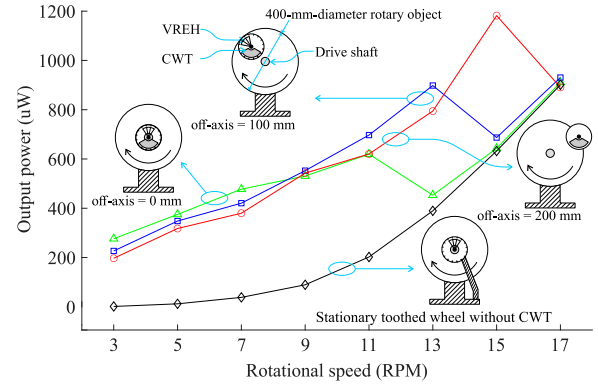


Figure 10: Output power of the energy harvesting system as a function of the rotational speed for different mounting positions.

several factors. In this study, we particularly investigate effects of the rotational speed and mounting displacement from the axis of rotation. As a reference, a harvester using an ordinary mechanical configuration with two anchor points is used, where the toothed wheel is secured to the stator side mechanically.

Figure 10 depicts the output power of the energy harvesting system using different mounting configurations as a function of the rotational speed. At low angular speeds (ranging from 3 rpm to 9 rpm), the three different locations of the energy harvester have no notable impact on the power output. However, it can clearly be observed that the single-anchor harvesters outperform the two-anchor reference. The reason for this is the pendulum effect that is introduced in the single-anchor solution as a result of the interaction between cogging torque and gravitational torque.

At increased rotational speeds, the cogging torque frequency increases and due to inertia the single-anchor VREH behaves like its two-anchor equivalent. It is obvious from Fig. 10 that the rotational speed at which this occurs is a function of the mounting position, which can be explained by changing centrifugal forces.

Including the interface circuit into the system, not all energy from the VREH is available to the wireless sensor system. Some of the harvested energy is lost in the power conversion, which is expressed by conversion efficiencies in the rectifier and PMIC stages. Moreover, the equivalent impedances of the interface circuit and the sensor system define the working point, and thus the power transfer, of the energy harvester. These effects can be incorporated into a single efficiency parameter η_{system} , describing the fraction of the theoretically extractable energy from a harvester that actually is made available to the load.

Figure 11 depicts the system efficiency η_{system} for the different system installations as a function of the rotational speed. It can be seen that the efficiency increases with increasing speeds, which can be attributed to higher output voltages from the VREH. Similarly, the single-anchor VREH achieves higher efficiencies than the two-anchor reference at low rotational speeds, which is a result of the increased voltages that originate from the pendulum effects. However, no clear relationship between the different mounting locations and the system efficiency can be observed.

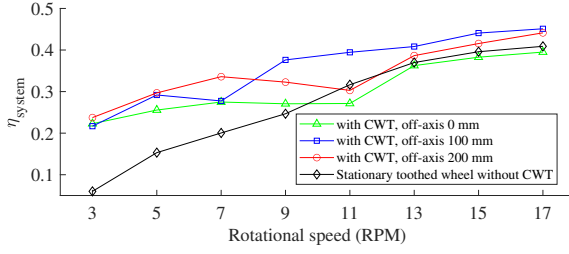


Figure 11: System efficiency (η_{system}) as a function of the rotational speed for different mounting positions.

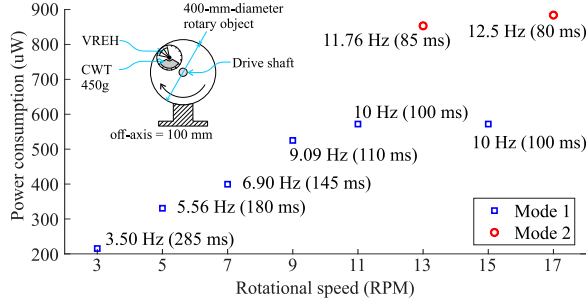


Figure 12: Attainable sample rate (sample interval) and power consumption of the wireless sensor system at different rotational speeds. Results are obtained for a system mounted at 100 mm off the axis of rotation.

A contributing factor to this may originate from the maximum power point tracking (MPPT) feature of the utilized PMIC (BQ25570). This MPPT operates periodically at a 16-second interval and adjusts the system to work at the maximum power transfer point based on open-circuit voltage. Due to the pendulum effect, transient voltages of significantly different amplitude may be detected by the MPPT circuitry, which may result in almost random and ineffective behavior of the MPPT functionality. As the function cannot be turned off, a potential solution may be to exchange the PMIC in further optimizations of the system.

Overall, system efficiency levels of approximately 10 to 50% are observable, demonstrating a potential for further performance improvements.

4.3 End-to-End System Evaluation

In order to evaluate the end-to-end performance, the energy harvesting system is employed to provide power to the wireless sensor. In this evaluation, the interface circuit, as depicted in Fig. 5, delivers a regulated voltage of 1.75 V through the PMIC's output. A short-term energy buffer is implemented by connecting a 700 μF capacitance to the storage terminal of the PMIC. Notably, a 30 μF capacitance would be sufficient for the sensor's duty-cycling operations. However, the 700 μF capacitance meets the additional energy requirement for power-on and initialization processes, ensuring a successful start-up of the system.

Figure 12 depicts the achievable sample rates that the wireless sensor system can be operated at under different rotational speeds.

The results in this figure exemplify the performance of a deployment with 100 mm distance to the axis of rotation. The sample rate increases as rotational speed increases, which corresponds to the increase in output power, which was observed earlier (cf. Fig. 10).

Through substantial pendulum effect at low rotational speeds, the wireless sensor can attain a sample rate of 3.5 Hz at a low speed of 3 rpm. However, as previously discussed, the weakening of the pendulum effect leads to a reduced output power at higher speeds. Consequently, at 15 rpm, the achievable sample rate is similar to that at 11 rpm. In addition, at both speeds of 11 rpm and 15 rpm, the wireless sensor could be operated in either Mode 1 or Mode 2. In Mode 1, the sensor reaches the maximum sample rate of 10 Hz due to BLE duty-cycling constraints discussed in Section 4.1. However, switching to mode 2 reduces the achievable sample rate, making mode 1 the preferred alternative. At 13 rpm and 17 rpm, sufficient power is generated to operate the sensor system in mode 2 with increased sample rate.

5 CONCLUSION

In this study, we designed, implemented and analyzed a self-powered, on-rotor sensor utilizing a single-anchor variable reluctance energy harvesting system. The wireless sensor system was designed to monitor the angular velocity of a low-speed, large-diameter rotary structure without being wired or battery-operated. The compact design, incorporating a cost-effective MEMS gyroscope and a robust electromagnetic energy harvesting system, makes it suitable for industrial and cost-sensitive applications.

The experimental evaluation demonstrates the feasibility of a VREH for enabling energy-autonomous operation of wireless sensors at low rotational speeds. With a low energy consumption of 60 μJ , the wireless sensor can complete a sample task, which includes sensor initialization, data acquisition, and wireless data transmission, resulting in an average power consumption of 75 μW at 1 Hz sample rate. By optimizing the energy consumption of the system in combination with an efficient harvester implementation, the sensor system can be operated at a sample rate of 3.5 Hz at very low rotational speeds of only 3 rpm.

Integrating a counterweight into the VREH not only enables the design of a single-anchor harvester with a gravity-supported stator, but also enhances the energy harvesting capabilities at low rotational speeds through pendulum effects. In conjunction with the operational mechanism of the gyroscopic sensor, this self-powered wireless sensing system realizes an RPM sensor that is easily installed directly on a rotating host. The contributions provided through this study demonstrate the efficacy and feasibility of utilizing a self-powered sensor system featuring a three-phase VREH with pendulum effects in practical applications.

Several areas remain to be investigated in further research studies. To enhance the system's performance, the counterweight configuration and magnetic coupling can be optimized, leading to a maximized pendulum motion. This involves analyzing factors such as counterweight mass, distance to the axis of rotation, and the geometric location of the center of mass. Furthermore, the energy consumption of the sensor system can be further reduced by hardware and software optimizations, such as including multiple sensor readings in the same data packet.

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